# YANCHEN LIU

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## EDUCATION HISTORY

**M.Sc. in Computer Science** **Sep. 2025 – Jun. 2027**
***University of Southern California***, LA, CA, US
**B.Eng. in Computer Science** **Sep. 2021 – Jun. 2025**
***Shanghai University***, Shanghai, CN

## RESEARCH EXPERIENCE

**Graduate Research Assistant** **Jun. 2025 - Present**
***University of Southern California*** - ***INK Lab*** *(Los Angeles, CA, US)*
**Topic**: Signals and Environments for Agentic RL; LLM Reasoning

1. Led the research of LLM-based game-playing agents, using replay-anchored judge as the long-horizon reward signal in reinforcement learning; Developed a text-based game-playing harness; Patched slime for training.
2. Led the research of user simulation analysis, designed a benchmark system for individual fidelity of user simulators; Proved that current methods are mixing individualization with population and recency.
3. Participated in the research of LLM-based recommendation agent; Proposed a new token-level self-distillation method for agentic RL on different recommendation tasks.
4. Proposed Soft-Thinking Policy Optimization, which explores the paradigm of LLM's latent-reasoning, including patching *vLLM* and *SGLang*, building an optimized Tree-Search generation and Training engine.
5. Enhanced *Chain-of-Thought* data with *segment and token importance* evaluated by different methods; Utilized the data for Supervised Fine-Tuning of various reasoning models with *unsloth*.

**Undergraduate Research Assistant** **Mar. 2023 - Apr. 2025**
***Shanghai University Cyber Security Lab*** *(Shanghai, CN)*
**Topic**: Vehicle Modeling, Simulation, and Intrusion Detection

1. Deep learning research on cyber security and functional safety in *in-vehicle communication system*.
2. Developed and optimized a novel *gradient descent solver* for *Multi-Dimensional Hawkes Process*; Introduced *MDHP-Net* for improved timing feature extraction in in-vehicle communication data.

**Undergraduate Research Intern** **Jun.2023 – Jul. 2024**
***Shanghai University*** & ***East China Air Traffic Control Bureau*** *(Shanghai, CN)*
**Topic**: Super Resolution for Meteorological Data

1. Deployed DL models including *PanGu* and *FourCastNet* to improve *meteorological monitoring, forecasting, and super-resolution* for the East China Air Traffic Control Bureau.
2. Utilized *key-point and semantic constraints*, combined with *feature map fusion*, to improve the up-sampling process in *super-resolution* models.

## INDUSTRY EXPERIENCE

**Software Development Engineer Intern, AI/ML** **Jun. 2026 – Aug. 2026**
***Amazon Lab 126*** - ***Kindle*** *(San Jose, CA, US)*
**Topic**: Benchmark System for Agentic Coding

1. Designed and built a benchmark system for Kindle coding agents. It auto-crawls the history of internal packages into a reproducible task dataset, benchmarks each agent configuration inside an execution harness, and reports performance metrics and analysis.
2. Designed the workflow as a three-stage pipeline: 1) sample generation (provide each sample with a leak-free spec plus a held-out hidden test), 2) agent run (develop on a clean-base tree behind a read/edit/build/self-test loop), and 3) evaluation (an oracle on f2p/p2p, an advisory LLM judge, and a behavioral-equivalence rescue).
3. Accelerated the system with a workspace pool and scheduler for isolation and parallel builds; Developed an anchor algorithm to reuse caches across different dependency snapshots, which turns hundreds of cold multi-minute builds into second-level warm rebuilds.

4. Built a knowledge-update evaluation gate for a GraphRAG knowledge base. Candidate memory and experience are evaluated by multi judges and coding harness, with human-in-the-loop for the final decision.

**Machine Learning Engineer Intern** **Jul. 2024 - Jun. 2025**

***Shanghai AI Laboratory*** *(Shanghai, CN)*

**Topic**: LLM Inference Engine, AI Compiler for Graph, Post-training

1. Extended *vLLM* for *LLM inference* on in-house TPUs, including researching and adapting *Speculative Decoding*, *Paged-Attention* and *Continuous-Batching*.
2. Developed *high-performance kernels* with *MLIR* to ensure seamless compatibility and optimal performance of LLMs on in-house TPUs.
3. Conducted research on *LLM knowledge injection and fine-tuning* for *kernel fusion and translation* across different hardware platforms.

## HONORS AND AWARDS

1. [2024] *First Prize* and *Group Competition Award* in 2024 ASC Student Supercomputer Challenge Global Final.
2. [2022] *First-Class Academic Scholarship* for outstanding academic performance, Shanghai University.

## PUBLICATIONS

† Equal contribution. * Corresponding author.

2026 ------

1. [ICLR Poster] S. Wang, **Y. Liu**, X. Ren. "*Segment-Level Attribution for Selective Learning of Long Reasoning Traces*".
2. **Y. Liu†**, X. Che†, B. Joshi†, R. Fukuda, J. Liu, R. Li, Y. Ge, H. Yan, X. Ren. "*Replay-Grounded Signals for Long-Horizon Civilization Agents*".
3. **Y. Liu†**, J. Liu†, B. Joshi†, X. Che, R. Li, X. Ren. "*Attribution Trials: Auditing Whether User Simulators Represent the Intended Individual*".
4. B. Joshi†, **Y. Liu†**, S. Swayamdipta, X. Ren, T. Paek, R. Koncel-Kedziorski. "*Learning to Seek User Information for Personalization via Multi-turn Conversations*".
5. J. Liu†, **Y. Liu†**, L. He, Z. Zhang, X. Che, L. Ge, F. Qian. "*When Repairs Stop Being Recipes: Re-Grounding Repair Memory for Coding Agents*".
6. [arXiv] X. Che†, H. Yan†, **Y. Liu†**, H. Liu, R. Li, A. Shi, H. Wang, J. Liu*. "*Does Learning to Predict the World Help Agents Act? Auditing World-Model Post-Training*".
7. [arXiv] X. Che†, Y. Ge†, S. Li†, **Y. Liu†**, H. Yan†, X. Lei†, Y. Wang, Z. Dong, Y. Yao, Q. Xie, L. Zhu, J. Liu*. "*GameLogicBench: Evaluating Coding Agents on Runtime Game Logic with Tick-Level State Assertions*".

2025 ------

8. [arXiv] J. Lv, X. He, **Y. Liu**, X. Dai, A. Shen, Y. Li, J. Hao, J. Ding, Y. Hu, S. Yin. "*HPCTransCompile: An AI Compiler Generated Dataset for High-Performance CUDA Transpilation and LLM Preliminary Exploration*".
9. [arXiv] Q. Liu†, **Y. Liu†**, R. Li, C. Cao, Y. Li*, X. Li*, P. Wang, R. Feng, S. Bu. "*MDHP-Net: Detecting an Emerging Time-Exciting Threat in IVN*".

2024 ------

10. [IEEE Internet of Things Journal] Q. Liu, X. Li, K. Sun, Y. Li*, **Y. Liu***. "*SISSA: Real-Time Monitoring of Hardware Functional Safety and Cybersecurity With In-Vehicle SOME/IP Ethernet Traffic*".
11. [MDPI Future Internet] X. Li, R. Li, **Y. Liu**. "*HP-LSTM: Hawkes Process-LSTM-Based Detection of DDoS Attack for In-Vehicle Network*".

## TECH STACK

**Programming Languages**

C++ (cmake, bazel, vcpkg, gcc, clang, msvc, msbuild, simd); CUDA (simt, nsight compute); Python (pip, conda, uv)

**AI/ML**

1. **Post-Training & (Agentic) RL** (lora, sft, opd, rlhf, ppo, grpo, dapo, opsd, sdpo, agentic rl; trl, verl, slime)
2. **Inference Acceleration** (TP, prefilling, decoding, prefix caching, continuous-batching, Chunked Prefill, Speculative Decoding, Quantization, Paged Attention, Flash Attention, MQA, GQA, MLA)
3. **Training Acceleration** (Mixed Precision, Gradient Checkpointing, ZeRO, DP, DDP, FSDP, MP, PP)